

Royco Dusk

Ankur Dubey
Waymont

Shivaansh Kapoor
Waymont

Jeff Bennett
Balancer Labs

Tomer Ganor
Certora

June 2026

Contents

1	Overview	2
1.1	Tranches and NAVs	2
1.2	The profit and loss waterfall	2
1.3	Dawn to Dusk	3
1.4	Components of a Dusk market	4
2	Notation	4
3	The fixed-point formulation	5
3.1	What we must prove	5
3.2	The sync map F	5
3.3	The BPT valuation P_A	6
3.4	The fixed-point equation	7
4	Existence and uniqueness of the solved rate	7
4.1	Monotonicity: E is non-increasing	8
4.2	Bracketing: the endpoints straddle the root	9
4.3	What the argument assumes about the pool	10
5	The on-chain solver	10
5.1	Integer bisection	10
5.2	Probe value reuse	10
5.3	The search bracket and the expressible range	12
5.4	Why the solve always converges	12
5.5	Iteration count and rounding	13
5.6	Implementation obligations	13
5.7	Rate-freshness obligations	14

1 Overview

This section offers a brief primer on Royco’s structuring substrates: Dawn and Dusk. Although this document specifies Dusk, the two substrates share their core machinery: tranches, the per-tranche NAVs, and the profit and loss waterfall.

1.1 Tranches and NAVs

A Royco market structures an asset into two complementary investment opportunities with distinct risk/return profiles called *tranches*:

- **Senior tranche (ST):** Protected capital. Lower returns for receiving first-loss coverage.
- **Junior tranche (JT):** Levered capital. Higher returns for providing first-loss coverage.

Moreover, each tranche holds a perpetual obligation to the other. The senior tranche pays the junior a share of its yield (risk premium) for absorbing losses first. The junior tranche, in turn, stands as a first-loss buffer: senior losses consume junior capital in full before the senior tranche incurs any. Accounting for these obligations requires distinguishing what each tranche *deposited* from what it is *entitled to*. Hence, Royco tracks two NAVs (net asset value) per tranche:

- **Raw NAV:** The mark-to-market value of the tranche’s *deposited* assets.
- **Effective NAV:** The value that each tranche *is entitled to* after its obligations to the other are settled.

1.2 The profit and loss waterfall

Every accounting synchronization marks the tranche NAVs to market in three steps:

1. **Measure:** Read the current raw NAVs and compute the delta against their last check-pointed values. Each delta represents the unrealized profit or loss of that tranche’s holdings since the previous accounting sync.
2. **Attribute:** Past coverage and risk premiums can leave a tranche entitled to more than its own holdings are worth. The excess ($ST_{\text{eff}} - ST_{\text{raw}}$ when positive, and symmetrically for JT) is a claim on the *other* tranche’s holdings. By conservation the two excesses are equal and opposite, so one tranche’s excess is exactly the other’s shortfall and the cross-claim runs one way. Each side’s raw NAV delta is split pro rata over the claims on it. The result is a pair of effective NAV deltas that sum to the total profit and loss.
3. **Settle:** The effective deltas are routed through the *PnL waterfall* in fixed priority order:
 - *Loss:* Utilize JT’s coverage buffer (JT_{eff}) before ST books any. Any coverage that JT advances to ST is booked as *JT impermanent loss* (ST’s debt to JT). Losses exceeding JT’s buffer reduce ST_{eff} and are booked as *ST impermanent loss* (JT’s debt to ST). JT IL is transient: it exists only during the market’s fixed-term regime, a window that lets the tranchised asset recover before the junior realizes losses.
 - *Recovery:* Repay any debts before gains accrue. ST’s IL has the first claim on appreciation from either side, and JT’s IL has the second claim on senior appreciation only.
 - *Yield:* Junior appreciation remaining after ST IL repayment accrues to JT in full. Senior appreciation remaining after all IL repayment is true yield, split between ST and JT at the rate set by the yield distribution model (YDM).

We denote the attribution and settlement steps as F throughout this document. Holding the checkpoint fixed, F maps the current raw NAVs to the new effective NAVs:

$$F: (ST_{\text{raw}}, JT_{\text{raw}}; \text{checkpoint}) \mapsto (ST_{\text{eff}}, JT_{\text{eff}}).$$

The semicolon separates variables from parameters: fixing the checkpoint yields a function of the raw NAVs alone. The checkpoint holds the previous sync’s raw NAVs, effective NAVs, outstanding ILs, market state, and yield-accrual bookkeeping (the time-weighted YDM accumulator and its timestamps). It is read from storage when the sync begins and cannot change while it runs. Every property stated for F holds for each fixed checkpoint.

F holds two key properties:

1. **NAV conservation:** the inputs sum to the outputs, $ST_{\text{raw}} + JT_{\text{raw}} = ST_{\text{eff}} + JT_{\text{eff}}$, for every checkpoint that itself conserves. Every stored checkpoint conserves, because it is itself a prior output of F . F redistributes value between the tranches without creating or destroying it.
2. **Piecewise-affine:** each branch is affine in its inputs, with slope $\in [0, 1]$. Combined with NAV conservation, this makes F continuous, non-decreasing, and 1-Lipschitz in each input. This is the property invoked in the existence proof (Section 4).

1.3 Dawn to Dusk

Dawn transforms the *risk profile* of any asset, but not its *liquidity profile*: senior shares have no exit path other than primary redemption.

Dusk extends the junior tranche to transform both. In Dawn, junior capital may sit in any priceable asset (Treasures, a money market fund, or co-investment alongside the senior tranche), provided its value is exogenous to the market itself. Dusk drops that restriction: junior capital is deployed into an AMM pool paired against the senior tranche’s shares.

- **Risk transformation:** Coverage still flows from JT to ST exactly as in Dawn: the pool position is JT’s first-loss collateral.
- **Liquidity transformation:** The pool position gives senior holders an instant secondary market, constantly quoting a price to buy and sell tranche shares for stables.

For the purposes of this document, the Dusk JT’s asset is a Balancer V3 Pool Token (BPT) whose constituents are the senior tranche share and a stable quote asset (USDC, sUSDS, or srRoyUSDC).

The recursion: Marking JT to market is no longer a plain oracle read. To value the BPT, the pool’s senior share leg must be priced at the senior rate $y = ST_{\text{eff}}/N$, where N is the total supply of senior shares. But ST_{eff} is not an observable quantity: it is the output of the sync map F , and F cannot run without its JT_{raw} input, which is the BPT value we set out to compute. The dependency chain closes on itself, as Figure 1 shows.

Economically, the junior tranche’s collateral contains the very senior shares its coverage protects. No evaluation order can resolve the cycle: the two unknowns, JT_{raw} and ST_{eff} (y is simply ST_{eff} per share), must be determined simultaneously. Under conditions made precise in Section 4, exactly one value of ST_{eff} satisfies both dependencies at once: price the BPT at its implied rate $y = ST_{\text{eff}}/N$, apply F , and the same ST_{eff} returns. This self-reproducing value is a *fixed point*. Section 3 formulates it, Section 4 proves it exists and is unique, and Section 5 finds it on-chain.

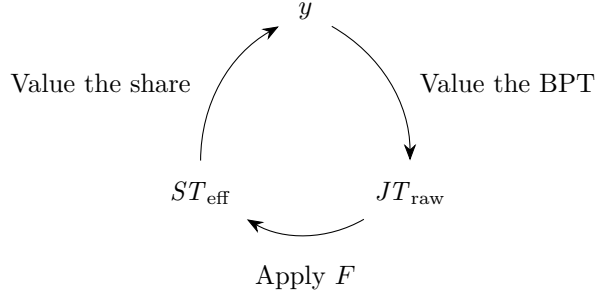


Figure 1: The recursion. Valuing the junior collateral needs the senior rate, and the senior rate depends on that valuation.

1.4 Components of a Dusk market

A Dusk market adds three components to the Dawn stack (pool, kernel, hook) and reuses the Dawn accountant. Each plays one role in resolving the recursion:

- **Gyro E-CLP pool:** A Balancer V3 elliptic concentrated liquidity pool over the senior share and quote asset. The senior leg is priced by the rate the kernel publishes, so the solved rate is what the pool uses to value senior shares.
- **Dusk kernel:** One contract, two paths. Write: the sync entrypoint solves the fixed point (Section 5), commits the accountant checkpoint, and caches the solved rate y^* . Read: the kernel reports the cached rate y^* to the pool instantly.
- **Balancer V3 Dusk hook:** Forwards Balancer’s before-operation callbacks (swap, add and remove liquidity) into the kernel’s sync entrypoint. Balancer reloads balances and rates after the hook so every operation executes at a fresh senior share rate.
- **Accountant:** Shared with Dawn. During a solve, the kernel calls its view-only preview repeatedly to evaluate F at candidate rates. After convergence, exactly one call commits the resultant checkpoint.

2 Notation

This section defines the notation used in subsequent sections.

Unless stated otherwise, every quantity is an 18-decimal fixed-point integer (WAD, scale 10^{18}): NAV amounts, rates, ratios, senior tranche shares, and BPT amounts alike. The single exception is raw quote-asset amounts, which keep their token’s native decimals (e.g. 10^6 for USDC).

Table 1: Mathematical notation used throughout this document.

Symbol	Meaning	Units
<i>Tranche NAVs and supply</i>		
ST_{raw}	Raw NAV of the senior tranche’s own assets	NAV
JT_{raw}	Raw NAV of the junior tranche’s BPT collateral, $JT_{\text{raw}} = P_A(y)$	NAV
ST_{eff}	Senior effective NAV after the waterfall	NAV
JT_{eff}	Junior effective NAV after the waterfall	NAV
y	Senior rate (NAV per senior share), the candidate $y = x/N$ fed into the solve	NAV/share
y^*	Published senior rate, $y^* = x^*/N = ST_{\text{eff}}/N$ at the fixed point	NAV/share
y_0	Cached senior rate, read once at solve start (fixes L and the band)	NAV/share

Symbol	Meaning	Units
x	Candidate senior NAV (the bisection probe), equal to the solved ST_{eff} only at the root x^*	NAV
N	Total senior-share supply (including any fee shares), frozen at solve start; the same N in the clamp and the rate	shares
<i>The fixed-point machinery</i>		
F	Sync map, $(ST_{\text{raw}}, JT_{\text{raw}}; \text{checkpoint}) \mapsto (ST_{\text{eff}}, JT_{\text{eff}})$	function
checkpoint	Previous sync's raw NAVs, effective NAVs, outstanding ILs, market state, and yield-accrual bookkeeping	—
P_A	Conservative BPT valuation, $JT_{\text{raw}} = P_A(y)$	function
G	Return map (loop output), $G(x) = F(P_A(x/N))$, so $E(x) = G(x) - x$	function
E	Error function, $E(x) = F(P_A(x/N)) - x$, the bisection target	function
g	Loop gain (round-trip amplification of x), $g = f \leq 1$ by the clamp at N	dimensionless
<i>Pool reserves and geometry (Gyro E-CLP)</i>		
b_q	Quote-leg reserve at the fair point	NAV
q	Quote-asset rate (1 for non-rate-bearing quote tokens)	NAV/quote
f	JT's fractional share of the total BPT supply	dimensionless
α, β	Lower / upper bounds on the senior rate the pool can quote (the E-CLP price band, distinct from Royco's correlation parameter)	NAV/share
L	E-CLP invariant value	—
V_{max}	Conservative pool value at the upper corner of the frozen curve	NAV
$V(y)$	Fair pool value at marginal price y (envelope-theorem valuation)	NAV
$D(y)$	The pool's demand for senior shares at price y (the fair reserve), $D(y) = V'(y)$	shares

3 The fixed-point formulation

Every pool operation triggers a sync, and every sync must publish one number: the senior share rate y . Section 1.3 showed why this is a fixed point, not an oracle read. This section makes it precise: one equation in one unknown, then two facts about one function.

3.1 What we must prove

Write the senior NAV the sync must produce as x . We will see that a self-consistent x is exactly a zero of the *error function*

$$E(x) = F(P_A(x/N)) - x,$$

where P_A values the junior collateral at a candidate rate and F is the accountant's sync map. The on-chain solver finds that zero by integer bisection, and bisection needs exactly two facts:

(Monotonicity) E never rises: each step holds or falls. So the values where $E \geq 0$ form an initial segment, the crossing is *unique*, and bisection knows which way to step.

(Bracketing) E is non-negative at the bottom of the search range and negative at the top. This makes the crossing *exist* and brackets it.

The rest of this section defines the three objects F , P_A , and E . Section 4 proves the two facts. Section 5 runs the solver. The argument rests on nothing else.

3.2 The sync map F

On each sync, the Dusk kernel calls the shared Royco accountant, whose sync map we denote F . Given the current raw NAVs of both tranches, F runs the attribution and settlement steps

of Section 1.2 against the stored checkpoint and returns the effective NAVs:

$$F: (ST_{\text{raw}}, JT_{\text{raw}}; \text{checkpoint}) \mapsto (ST_{\text{eff}}, JT_{\text{eff}}).$$

Two properties of F matter downstream (Section 1.2). It conserves NAV ($ST_{\text{raw}} + JT_{\text{raw}} = ST_{\text{eff}} + JT_{\text{eff}}$), and it is non-decreasing and 1-Lipschitz in each input (Lemma 4.2). Conservation holds by construction: every stored checkpoint is itself a prior output of F .

In a Dusk market the senior raw NAV is exogenous to the loop: it is the value of the senior tranche's own assets, marked independently of the pool. Only the junior raw NAV depends on the senior rate, because the junior's collateral is the BPT. The senior side therefore drops out, and for one sync, with ST_{raw} and the checkpoint fixed, we read F as a function of a single variable:

$$F: JT_{\text{raw}} \mapsto ST_{\text{eff}}.$$

F returns ST_{eff} *gross* of protocol fees: a fee accrues as a separate NAV claim settled outside F , never netted out of ST_{eff} . So $y = ST_{\text{eff}}/N$ is consistent only when N counts the same shares the gross NAV backs, including any minted for fees.

3.3 The BPT valuation P_A

The junior tranche owns a fraction f of a Gyro E-CLP pool whose two legs are the senior share and a quote asset. $P_A(y)$ is the value of that position when the senior share trades at rate y , marked conservatively so that no manipulation and no rounding can ever credit the junior tranche more than it could withdraw.

The frozen curve. The senior leg is priced by the rate the kernel publishes. At the start of a solve the kernel reads the cached rate y_0 once and computes the pool invariant L from the rate-scaled balances, rounded down in favour of the senior tranche. L is then held fixed while the candidate rate y varies. Freezing L is what makes P_A a clean function of y alone, which the monotonicity proof relies on. The implementation details of the scaling and rounding are collected in Section 5.6.

The fair point. Given L , the conservative value at candidate rate y is the pool's value at its *fair point*: the point on the frozen curve whose *marginal price* (its local slope, the spot rate for an infinitesimal trade) equals y . Write this value $V(y)$ and write

$$D(y) = V'(y),$$

the pool's *demand* for senior shares at price y , that is, the senior reserve the fair point holds. Two geometric facts about the E-CLP, proved in Section 4.1, are all we need: V is non-decreasing, with $V'(y) = D(y) \geq 0$ (a higher senior price raises the value of the senior shares the pool holds), and the demand $D(y)$ is finite at every rate the curve can express.

The clamp. Far below the operating rate the curve's demand $D(y)$ grows without bound, but the pool can hold at most the total supply N (even if every senior holder sold into it). The true demand cannot exceed N , so we cap it there:

$$\hat{D}(y) = \min(D(y), N), \quad \hat{V}(y) = \int_0^y \hat{D}(u) du.$$

By definition $\hat{V}(0) = 0$, and $V(0) \geq 0$ since a reserve is non-negative. With $\hat{D} \leq D = V'$ this gives $0 \leq \hat{V} \leq V$ everywhere: the cap only lowers the mark and preserves conservatism. At any healthy rate the live demand is far below N and the cap is inactive. Its purpose is to bound demand by the supply, which Section 4 turns into unconditional monotonicity.

The valuation. The junior tranche’s pro-rata claim, rounded down in favour of the senior tranche, is the *conservative BPT valuation*:

$$P_A(y) = \lfloor f \cdot \widehat{V}(y) \rfloor. \quad (1)$$

By the geometric facts and the clamp, P_A is non-decreasing in y and Lipschitz with constant fN (Section 4.1). Within a single solve V depends only on L and y , not on the live pool composition. A fee-free swap conserves the pool’s true invariant, and the L the solver consumes is a round-down underestimate of it recomputed from live balances, so $L \leq L_{\text{true}}$ for every reachable composition. A swap (or a donation, which Balancer V3 ignores since it accounts internally) can therefore move the mark only within the round-down band, and never above the true fair value. The mark is manipulation-bounded, not immovable.

A capped lower bound, stated plainly. P_A is deliberately a lower bound on the junior’s realisable worth, not the pool’s live fair value. The two differ by exactly the value the cap discards,

$$V(y) - \widehat{V}(y) = V(0) + \int_0^y (D(u) - N)^+ du,$$

the phantom coverage resting on senior shares beyond the N that exist, plus any reserve held at rate 0. The gap is strict precisely when the clamp binds somewhere below y (or $V(0) > 0$); in a pool whose demand never reaches N on $[0, y]$ it vanishes and $\widehat{V} = V$ there. The under-marking is thus targeted: in the healthy band the cap is inactive and $\widehat{V}' = D$ matches fair value at the *margin*, while the *level* stays a lower bound because the low-rate phantom area was already excluded.

This is one of three independent senses the word *conservative* carries in this document; a valuation can have any one without the others.

Manipulation-bounded. The mark reads the frozen invariant L and the fair point, not live pool balances, so trading moves it only within the round-down band above.

Senior-favouring. Every rounding step — the floor in P_A , the round-down of L , the published rate toward senior — breaks ties for the protected tranche.

Capped lower bound. The clamp $\min(D, N)$ under-marks in the danger zone, by exactly the gap above.

The published senior rate y^* is built on this capped lower-bound valuation by design. Section 4 turns the same cap into unconditional monotonicity.

3.4 The fixed-point equation

Chaining the definitions, $JT_{\text{raw}} = P_A(ST_{\text{eff}}/N)$ and $ST_{\text{eff}} = F(JT_{\text{raw}})$, so the senior NAV must reproduce itself:

$$x = F(P_A(x/N)). \quad (2)$$

A solution is a zero of the error function E (Section 3.1), and the published rate is $y^* = x^*/N$. Section 4 proves E has exactly one zero (Monotonicity) inside a range whose endpoints straddle it (Bracketing).

4 Existence and uniqueness of the solved rate

This section proves the two facts from Section 3.1: E is non-increasing (Monotonicity), and its sign changes across a search range whose endpoints we can write down (Bracketing). Together they give a unique self-consistent rate that bisection finds deterministically.

Theorem 4.1 (the solved rate exists and is unique). *Fix ST_{raw} , the checkpoint, the supply $N \geq 1$, and the frozen invariant L at the start of a sync. On the bracket*

$$\mathcal{B} = [0, x_{\max}], \quad x_{\max} = ST_{\text{raw}} + \lfloor f V_{\max} \rfloor + 1,$$

where $V_{\max} = \sup_y V(y)$ is the pool value at the all-quote corner (the senior shares all sold), the integer error function $E(x) = F(P_A(x/N)) - x$ is non-increasing (Monotonicity) and satisfies $E(0) \geq 0$ and $E(x_{\max}) < 0$ (Bracketing). Hence the integers where $E \geq 0$ form an initial segment $[0, x^*]$ with a unique right endpoint x^* , the self-consistent senior NAV, and bisection on the predicate $E(x) \geq 0$ finds it.

Sections 4.1 and 4.2 prove the two pillars, and Section 4.3 records what the argument assumes about the pool, so an auditor can check it per market.

4.1 Monotonicity: E is non-increasing

$E(x) = F(P_A(x/N)) - x$. The $-x$ term falls by one per step, so it is enough to show the composition $x \mapsto F(P_A(x/N))$ rises by at most one per step. We bound each stage in turn.

Stage 1: $f\hat{V}$ moves by at most 1 per step, so P_A moves by 0 or 1. Fix the invariant L at the pre-solve cached rate, as Section 3.3 requires. The conservative pool value at rate y is the quote it holds plus its senior shares marked at y ,

$$V(y) = b_q(y) + y D(y),$$

where $D(y)$ is the pool's demand for senior shares at price y and $b_q(y)$ is the quote reserve at the same fair point. Moving along the curve $db_q/dD = -y$ (the slope of the reserve curve is the price), so the rebalancing terms cancel:

$$\frac{dV}{dy} = \underbrace{\frac{db_q}{dD} \frac{dD}{dy} + y \frac{dD}{dy}}_{=0} + D(y) = D(y) \geq 0.$$

This is the envelope theorem: the first-order effect of the rebalancing variable vanishes at the fair point, leaving the direct term. The cancellation holds on the open band (α, β) . Below it the pool is all senior and $V(y) = y D(\alpha)$ rises linearly; above it the pool is all quote and V is constant at $V_{\max}$. Both pieces are non-decreasing and join the band continuously, so V is non-decreasing throughout.

The clamp now does its one job. With $\hat{D}(y) = \min(D(y), N)$ the capped value $\hat{V}$ has $\hat{V}'(y) = \hat{D}(y) \leq N$ everywhere, so for $y > y'$,

$$0 \leq \hat{V}(y) - \hat{V}(y') \leq N(y - y').$$

The candidate rate is $y = x/N$, so a unit step in x moves $f\hat{V}$ by at most $fN \cdot \frac{1}{N} = f \leq 1$. A real increase of at most 1 changes a floor by 0 or 1, so $P_A = \lfloor f\hat{V} \rfloor$ rises by 0 or 1 per step, and never falls.

Stage 2: F is 1-Lipschitz. This is Lemma 4.2 below: $0 \leq F(j) - F(j') \leq j - j'$ for $j > j'$.

Composition. The chain $x \mapsto x/N \mapsto P_A \mapsto F$ multiplies the per-stage factors,

$$\underbrace{\frac{1}{N}}_{\div N} \cdot \underbrace{fN}_{P_A} \cdot \underbrace{1}_F = f \leq 1,$$

since the junior owns at most the whole pool ($f \leq 1$). This product is the *loop gain* $g = f$: the dollars that return to the senior NAV after one trip around the loop, per dollar sent in. Capping demand at the supply holds it at or below one *by construction*, with no runtime condition to check.

On the integer lattice this gives the result directly. A unit step in x moves P_A by 0 or 1 (Stage 1), and F is 1-Lipschitz (Stage 2), so $F(P_A(x/N))$ rises by at most 1. Subtracting the $-x$ term,

$$E(x+1) - E(x) \in \{-1, 0\}.$$

E is non-increasing, so the integers with $E \geq 0$ form an initial segment $[0, x^*]$. At its right endpoint the next step must be a drop of 1 (a flat step would keep $E \geq 0$), so x^* is the unique integer with $E(x^*) \geq 0 > E(x^*+1)$. The drop being exactly 1 and E integer-valued force $E(x^*) = 0$, an exact fixed point. Bisection returns this x^* , the largest self-consistent senior NAV. This proves Monotonicity.

Lemma 4.2 (F is non-decreasing and 1-Lipschitz in JT_{raw}). *Fix ST_{raw} and the checkpoint and vary $j = JT_{\text{raw}}$. Then $0 \leq F(j) - F(j') \leq j - j'$ for $j > j'$.*

Proof. F reaches ST_{eff} through two stages (Section 1.2), each with non-negative coefficients in $[0, 1]$: attribution splits the raw delta by a claim fraction $a \in [0, 1]$ fixed at the checkpoint (each tranche's claim on a leg is at most that leg's raw NAV, by conservation, so $a \leq 1$), and settlement routes the result through chains of min, max, addition and subtraction (slopes $\{0, 1\}$) and a yield split (fractions $w, 1 - w$ with w capped). So ST_{eff} is non-decreasing in j , and so is JT_{eff} . NAV conservation finishes it: $ST_{\text{eff}} + JT_{\text{eff}} = ST_{\text{raw}} + j$, so the two slopes sum to exactly 1, and since both are non-negative each lies in $[0, 1]$. Thus F is non-decreasing and 1-Lipschitz. $\square$

Remark (Uniqueness and the loop gain at $f = 1$). The clamp fixes the loop gain at $g = f \leq 1$ (Section 4.1), which makes E non-increasing but permits it to be flat; the two regimes part at the boundary. When $f < 1$ strictly, E loses a full wei of altitude every $1/(1 - f)$ steps on average, so the root is unique — in real arithmetic, and within a wei of it on the lattice. When $f = 1$ the return $F(P_A(x/N))$ can track x exactly, and the zero set $\{x : E(x) = 0\}$ widens into a contiguous integer block: a *plateau* of self-consistent NAVs in place of a single root. Strict uniqueness therefore carries the hypothesis $f < 1$, which the clamp alone does not supply.

This costs nothing, because the published value is *defined* to be the largest plateau integer, the x^* of Lemma 4.1, and that choice is immaterial to the junior tranche. Across the block $E = 0$ gives $ST_{\text{eff}} = x$, so by NAV conservation $JT_{\text{eff}} = ST_{\text{raw}} + P_A(x/N) - x$. A unit step inside the block raises $F(P_A(x/N))$ by exactly 1 (the $-x$ term fell by 1 and E held at 0); as F is 1-Lipschitz and P_A steps by 0 or 1 (Section 4.1), this forces P_A up by exactly 1, whence

$$JT_{\text{eff}}(x+1) - JT_{\text{eff}}(x) = (P_A((x+1)/N) - P_A(x/N)) - 1 = 1 - 1 = 0.$$

The junior's entitlement is identical at every plateau point. This invariance is exactly as strong as F 's integer conservation: it relies on $ST_{\text{eff}} + JT_{\text{eff}} = ST_{\text{raw}} + JT_{\text{raw}}$ holding at wei precision, the two outputs not independently rounded — which the accountant guarantees by construction (Section 1.2). Existence does not degrade at $f = 1$ either: the $+1$ slack in x_{max} (Section 4.2) makes $E(x_{\text{max}}) \leq -1 < 0$ whatever the gain, so the bracket still straddles a crossing. Strict uniqueness needs $f < 1$; safety needs only the clamp.

4.2 Bracketing: the endpoints straddle the root

We exhibit a range whose endpoints have opposite signs by construction, so no runtime sign-change check can fail.

Lower endpoint. At $x = 0$, $E(0) = F(P_A(0)) = ST_{\text{eff}} \geq 0$, since the accountant holds effective NAVs non-negative, whatever the junior collateral is worth.

Upper endpoint. P_A is bounded above by $\lfloor fV_{\text{max}} \rfloor$. Since V is non-decreasing, $\hat{V} \leq V \leq V_{\text{max}}$ for every candidate rate, whichever way the pool sorts its tokens. By NAV conservation $ST_{\text{eff}} = ST_{\text{raw}} + JT_{\text{raw}} - JT_{\text{eff}} \leq ST_{\text{raw}} + JT_{\text{raw}} \leq ST_{\text{raw}} + \lfloor fV_{\text{max}} \rfloor$ (using $JT_{\text{eff}} \geq 0$) for every candidate. At $x_{\text{max}} = ST_{\text{raw}} + \lfloor fV_{\text{max}} \rfloor + 1$ we therefore have $E(x_{\text{max}}) = ST_{\text{eff}} - x_{\text{max}} \leq -1 < 0$.

Both endpoints are computed once, at solve start, from frozen values. The sign change is a theorem, not a checked precondition.

4.3 What the argument assumes about the pool

The proof uses only two structural facts, which an auditor verifies once per market:

Waterfall. F is non-decreasing and 1-Lipschitz in JT_{raw} (Lemma 4.2). This is a property of the shared accountant and holds for any Royco market using the standard waterfall.

Pool. The fair value V is non-decreasing in y , with $V' = D \geq 0$ by the envelope theorem (Section 4.1). This holds for the Gyro E-CLP and for any constant-function pool.

The proof never differentiates F , so it survives the waterfall’s kinks, and never inverts a pool invariant, so it transfers unchanged across pool types: only the demand D that the clamp caps is pool-specific.

Out of scope. The solve fixes a manipulation-bounded rate; it does not monitor the live pool’s *composition*. As senior shares accumulate and quote drains, the published mark stays correct — it reads the frozen curve, not live balances — but tracking that drift is a separate operational concern. So is Royco’s coverage and correlation parameter, a distinct quantity from the pool’s upper price bound β (Table 1); neither is addressed by the fixed point.

5 The on-chain solver

Lemma 4.1 gives a unique root in a bracket whose endpoints straddle it. This section runs the solver: integer bisection, in bounded gas. The conditions that make the theorem apply on-chain are stated as explicit obligations.

5.1 Integer bisection

The solver bisects E on $\mathcal{B}$ at integer (WAD) precision. Each step halves the bracket and keeps the half whose endpoints still straddle the sign change. The bracket holds $|\mathcal{B}| = x_{\text{max}} + 1$ integers, so after $K = \lceil \log_2(|\mathcal{B}|) \rceil$ steps it has collapsed to a single integer x^* , the boundary with $E(x^*) \geq 0 > E(x^* + 1)$. The published rate is $y^* = x^*/N$, rounded toward the senior tranche.

Bisection, not Newton, for three reasons. F is non-smooth, its slope jumping at branch boundaries (Lemma 4.2), so a tangent-following method has no derivative to follow, whereas bisection needs only that E be non-increasing. The iteration count K is fixed at deploy time, so gas is constant with no data-dependent exit. And the loop is a short sweep over already-audited valuation code, with no invariant inversion and a unique root.

5.2 Probe value reuse

Write the loop output as $G(x) = F(P_A(x/N))$, so $E(x) = G(x) - x$. To get the sign of $E(z)$ the solver already evaluates $G(z)$ in full — it prices the junior collateral at the candidate rate

and runs the waterfall — then keeps one bit and discards the number. But that number is not merely a test statistic. Monotonicity (Section 4.1) certifies $G(z)$ as a bracket endpoint on the correct side of the root, usually far tighter than the midpoint rule reaches in many steps.

Lemma 5.1 (probe upgrade). *Let z be a probe and $v = E(z)$, so the loop output $G(z) = z + v$ is already in hand. If $v \geq 0$ then $z \leq x^*$ and $G(z) \in [z, x^*]$; if $v < 0$ then $z > x^*$ and $G(z) \in [x^*, z - 1]$. Either way $G(z)$ is a valid endpoint on z 's side of the root, no farther from x^* than z .*

Proof. By Monotonicity (Section 4.1) the integers with $E \geq 0$ are exactly $[0, x^*]$, so the sign of v locates z relative to x^* . And $G = F \circ P_A$ is non-decreasing, both factors being so (Section 4.1 and Lemma 4.2). If $z \leq x^*$: $G(z) \leq G(x^*) = x^*$ and $G(z) = z + v \geq z$. If $z > x^*$: $G(z) \geq G(x^*) = x^*$ and $G(z) = z + v \leq z - 1$, since $v \leq -1$. $\square$

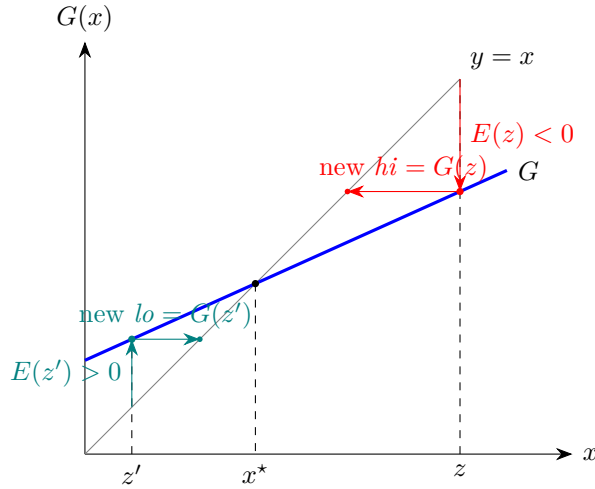


Figure 2: Probe value reuse, one step. The root x^* is where the return map G meets the diagonal $y = x$. A probe at z right of the root lands the curve point $(z, G(z))$ below the diagonal yet no lower than x^* (Lemma 5.1); sliding horizontally to the diagonal makes $G(z)$ the new upper endpoint, already past the midpoint and toward the root. A probe z' left of the root gives a new lower endpoint symmetrically. On a flat stretch of G — a covered, locally inert market state — the first slide lands at the root's level at once: the 2–3 probe solve of Lemma 5.2.

The solver applies the lemma at every step: where plain bisection moves the endpoint to the midpoint m , value reuse moves it to the value the midpoint returned, $G(m) = m + v$. Maintaining $x^* \in [lo, hi]$ from the bracket of Lemma 4.1:

```

( $lo, hi$ )  $\leftarrow$  ( $0, x_{\max} - 1$ )
while  $hi > lo$  :
     $m \leftarrow lo + \lceil (hi - lo)/2 \rceil$ 
     $v \leftarrow E(m)$     (one probe;  $G(m) = m + v$ )
    if  $v \geq 0$  :  $lo \leftarrow m + v$   else :  $hi \leftarrow m + v$ 
return  $lo$ 

```

Proposition 5.2 (value reuse is free). *The loop maintains $x^* \in [lo, hi]$, terminates in at most K probes — the same worst case as Section 5.1 — and returns the bit-identical x^* , including the largest-integer selection on the $f = 1$ plateau (Section 4.1). The published rate, the committed checkpoint, and the obligations of Section 5.6 are all unchanged.*

Proof. The invariant holds initially by Bracketing (Section 4.2); each update replaces an endpoint by $G(m)$, which Lemma 5.1 places between that endpoint and x^* inclusive. Progress: $lo < m \leq hi$, so the $v \geq 0$ branch gives $lo' = m + v \geq m > lo$ and the $v < 0$ branch gives $hi' = m + v \leq m - 1 < hi$ — never worse than moving to the midpoint, so the $\lceil \log_2 \rceil$ bound is inherited and reuse can only beat it. At termination $lo = hi = x^*$. On the $f = 1$ plateau the $v \geq 0$ branch pushes lo upward through the block ($lo' = G(m) \geq m$), so the returned point is the largest integer with $E \geq 0$, the x^* of Lemma 4.1. The only change from Section 5.1 is the bracket-update rule. $\square$

The gain is large because the recursion is locally inert wherever the junior buffer fully covers the senior delta ($\partial ST_{\text{eff}}/\partial JT_{\text{raw}} = 0$): one value jump lands on the root, so healthy and stressed-but-covered states — including $f = 1$ and a binding clamp — solve in 2–3 probes. The loop is live only where the senior holds claims on the pool leg (IL recovery, cross-claim attribution, the yield split), and there each jump still shrinks the overshoot by the local gain, so even deep distress converges in ~ 20 probes against the ~ 64 of width-limited bisection. Since the solve runs on every swap, add, and remove, this is a 5–30 $\times$ cut in solver gas on the hot path, at the cost of a single changed line and no new assumptions.

5.3 The search bracket and the expressible range

The solver uses the *conservation bracket* $\mathcal{B}$ of Lemma 4.1 as its search domain. Its endpoints straddle the root by construction, so the search always brackets a unique answer. Each probe evaluates the frozen- L valuation in closed form, not the pool’s invariant solver, so no probe re-triggers the E-CLP domain check. That check runs once, when L is computed at solve start from the live balances.

The root is the economically correct rate, but the deployed pool can only *quote* rates inside its price band. Write α and β for the lower and upper bounds on the senior rate $y = x/N$ after the mapping below; the expressible range is then

$$x \in [N\alpha, N\beta]. \quad (3)$$

These bounds are per-solve. The pool’s configured band constrains the *rate-scaled* senior price, so it is recentered at the cached rate y_0 , and its orientation depends on the token sort:

- if the senior share is token0 (price quoted as quote-per-senior), the senior-rate bounds are the configured bounds directly;
- if the senior share is token1 (price quoted as senior-per-quote), they are the reciprocals of the configured bounds, swapped, then recentered at y_0 ; at rates near 1 this reduces to the $[N/\beta, N/\alpha]$ form of earlier materials.

After solving, the kernel checks $x^* \in [N\alpha, N\beta]$. If the root falls outside, the rate is real but unquotable: the sync reverts rather than clipping to a band edge and publishing a rate the pool cannot honour. This is the only market-condition revert in the solve, and it is a genuine economic signal (a senior loss larger than the pool’s band can represent), not a solver fragility.

5.4 Why the solve always converges

Lemma 4.1 is a statement about the function E . For it to govern the on-chain loop, three implementation invariants must hold. Each is cheap and is listed as an obligation in Section 5.6.

Frozen inputs. Every input to E (ST_{raw} , N , f , the invariant L , the cached rate y_0 , and the checkpoint) is read once at solve start and never re-read. Bisection compares E at different probes and assumes one fixed function. Any external reads inside F (the yield-model view,

the block timestamp) are constant across the bisection, since it runs in one block against a frozen checkpoint, so no probe sees a different function.

Live supply. $N \geq 1$. The map divides by N , so the supply must be positive. This is a *Dusk obligation*, not a property inherited from the shared tranche. The Dawn senior tranche mints shares only on deposit and burns them on redemption, seeds nothing at market creation, and applies no virtual-share offset in its conversion (at zero supply it converts one-to-one), so its supply can fall back to zero. A Dusk market must therefore pin $N \geq 1$ for the life of the market by construction — for instance a non-redeemable minimum senior position seeded at creation, or a virtual-share offset in the senior conversion — so the division is always defined. The same floor is what discharges the $N \geq 1$ hypothesis of Lemma 4.1.

Total E . Every probe returns a value rather than reverting. E has two parts. P_A maps every $x \in \mathcal{B}$ to a senior balance inside the E-CLP’s computable domain (§5.3), so it never reverts. F reverts only on a checkpoint that violates NAV conservation, and every checkpoint the accountant commits conserves by construction (Section 1.2). So on the reachable state space E is defined at every probe in $\mathcal{B}$.

Under these three invariants, the solve terminates in exactly K steps and returns the unique root. The only remaining reverts are external to the solver: a paused market, a stale or reverting price oracle, or the surrounding pool operation hitting its own Balancer limits. None of these is a failure of the solve, and in each case the cached rate is left untouched. Two guarantees follow, and only these two: the failed operation reverts before it can *consume* a stale rate, and a failed sync never *overwrites* the cache. Ordinary view reads (`getRate`) still return the last published rate — staleness is prevented from propagating through an operation, not from being read.

5.5 Iteration count and rounding

Two policy choices fix the solver’s behaviour:

Iteration count is the K of Section 5.1, hardcoded per market. Each step of E lies in $\{-1, 0\}$ (Section 4.1), so K halvings collapse the bracket to the single integer boundary x^* . No residual or iteration-count check is needed.

Rounding sends the final x^* , and the published $y^* = x^*/N$, toward the senior tranche. Together with the floor inside P_A (Eq. (1)) and the round-down of the invariant, every rounding step favours the protected tranche.

5.6 Implementation obligations

The correctness argument rests on the following, each verifiable per market. The first group makes E the right function, the second makes the solve converge, the third keeps every step conservative.

Senior independence. The senior raw NAV is marked from sources that do not read the pool (its BPT value, its spot price, or the published rate), so the loop has exactly one fixed-point variable.

Valuation wiring. The senior leg is lifted to WAD by its decimal factor 10^{18-d_s} (d_s its native decimals) and the published senior rate. The quote leg is lifted by 10^{18-d_q} and its rate q .

Frozen curve. L is computed once per solve from the balances scaled at y_0 , rounded down (the underestimate), and held fixed across the bisection. $V_{\max}$ is the value at the upper corner of this same frozen curve.

Clamp. The valuation uses the integral form $\widehat{V}(y) = \int_0^y \min(D, N)$ (Section 3.3), with N read at solve start. In closed form this is Ny below the rate where demand falls to N , and V shifted by a constant above it (the split is unique when demand is monotone, as for the Gyro E-CLP). Capping the share count inside the point value, $b_q + y \min(D, N)$, is a different function that is not N -Lipschitz and breaks monotonicity.

Frozen inputs, live supply, total E . As in Section 5.4: read once, $N \geq 1$, conserving checkpoint.

Conservative rounding. Senior-favouring at every step: the invariant down, the valuation floor, and the published rate toward senior.

5.7 Rate-freshness obligations

The solver publishes y^* ; the surrounding pool must consume it freshly. These obligations are on the hook–kernel–pool wiring rather than the solve, and are listed for the implementer.

Sync coverage. The hook registers all three before-operation callbacks — before-swap, before-add, and before-remove — so every operation triggers a sync.

Reload ordering. The kernel syncs inside the matching before branch; Balancer reloads balances and rates for that operation only after the hook returns, so the operation executes against the freshly published rate. The reload reads the Vault’s own internal balances, so the hook must publish through a Vault entrypoint — a raw token transfer into the pool is not seen by the reload.

Non-reverting view. `getRate` returns the cached y^* and must not revert merely because the rate looks stale.

Failure semantics. A failed sync reverts the operation and does not overwrite the cache; ordinary view reads still return the last published rate (Section 5.4).

Recovery mode. Recovery-mode removes are a separate path, not bound by this contract.